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Summer School 26
Iot & Drone Building

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RoLL No :- 41

Drone Assignment

Date :- 2 August 26.

Q1 What is a drone (UAV)? explain its definition, history and evolution.

Ans A Drone UAV (unmanned aerial vehicle) is essentially a flying robot that operates without a human pilot on board.
Instead of sitting inside a cockpit, the pilot controls the aircraft remotely from the ground, or the drone flies autonomously using pre-programmed GPS routes, onboard sensors and artificial intelligence.

Key components of a UAV

→ The Airframe : The physical body, wings or rotors that keep the drone airborne.

→ Flight controller : The "brain" of the drone containing gyroscope and accelerometers to maintain balance and stability.

→ Propulsion system : Motors and propellers powered by batteries or fuel.

> **payload** :- The equipment the drone carries, such as cameras, thermal sensors or delivery packages.

History and Evolution

1. **Early 1900s** (The start)

During world war I and II, inventors created the first remote-controlled flying machines. They were mostly used as flying targets to practice shooting down enemy planes.

2. Late 1900s (The spy era):

By the vietnam war, armies started putting cameras on drones to fly over dangerous enemy territory and take photos without risking pilot lives.

3. 2000s (The GPS feedback upgrade):

With the arrival of GPS and digital tech, drones became much smarter, allowing soldiers to control them, from thousands of miles away

4 2010 to present (The Everyday Boom)

As smartphone parts, batteries and sensors became smaller and cheaper, drones entered the consumer market.
Today, anyone can fly a drone for fun, take wedding photos, inspect farm crops, or deliver packages

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Q2 Differentiate between

a) UAV & Drone

UAV	Drone
a) Def :- A precise, tech and aviation-industry term defined by regulatory bodies (like the FAA or military)	Def :- A broad, casual, everyday term used by the general public to describe any flying gadget without a human pilot
b) scope and Inclusions	scope and Inclusions
Refers strictly to the flying aircraft itself	often used as an umbrella term that includes not just the aircraft, but

	the entire ecosystem including the ground control station, software, cameras, and communication tch links
c) origin of the word	c) origin of the word
Born out of aviation engineering, legal frameworks, and defense nomenclature to officially classify pilotless aircraft	originally borrowed from the military nickname for target practice aircraft
d) context of usage	d) context of usage
Primarily used in professional, legal, scientific and military documentation	used in pop culture, consumer marketing; media and hobbyist circles.
eg: commercial UAV regulations").	e.g, "buying a camera drone for vacation".
e) Level of autonomy & complexity	e) Level of autonomy & complexity.
usually implies a more sophisticated highly engineered	can refer to anything simple and small - from a cheap toy mini

aircraft capable of complex, pre-programmed flight paths used for enterprise or defense.	mini-quadcopter you fly in your living room to a high end cinematic filming rig

B) Diff b/w Fixed-wing & Multi rotor Drone

Feature	~~Fi~~ Fixed-wing	Multi-Rotor
a) Design	Traditional airplane shape with rigid wings; uses aerodynamics for lift	Helicopter or quadcopter style with multiple spinning rotors
b) Takeoff & Landing	Require a runway, launcher, or hand launch; lands via glide, belly flop or parachute	VTOL (vertical takeoff and Landing); can lift off and land straight up and down.
c) Hovering Ability	cannot hover, must stay in constant forward motion to stay airborne	can hover in one fixed spot and execute precise maneuvers.
d) speed & Range	high speed capable of covering massive distances	slower speed with restricted range

e)	Flight Duration	Long flight times due to energy efficient wing fit	short flight times typically limited to 20-40 minutes, due to heavy battery drain

C) Diff b/w Quadcopter & Hexacopter

	Feature	Quadcopter	Hexacopter
a)	Number of motors.	Exactly 4 motors and propellers	exactly 6 motors and propellers.
b)	Payload capacity	Lower lifting capacity, best for lightweight cameras and sensors	High lifting capacity, can carry heavy professional cinema cameras or industrial Payloads
c)	wind Resistance & stability	good stability in moderate winds, but can struggle in heavy gusts	excellent stability and wind resistance due to the extra thrust and control authority.

d)	safety & Redundancy	Lower redundancy. If a single motor or propeller fails, the drone will almost certainly crash	High redundancy. If a one motor fails mid-flight, the flight controller can often compensate allowing the drone to land safely.
e)	cost and complexity	cheaper to build, repair & maintain	More expensive, complex to main

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Q3. List at Least five to Real life example/application of drones

1. Agricultural surveying and crop monitoring

- How it's used: Farmers use multispectral drones to fly over large fields and monitor crop health. The cameras capture data on plant growth, moisture levels and nitrogen status, helping farmers apply fertilizers and water precisely where needed to save costs and boost yields.

2. Infrastructure & Pipeline Inspection

Drones safely inspect high risk, hard to reach infrastructure such as wind turbine blades, cell towers, oil and gas pipeline and massive bridges. Equipped with high-resolution and thermal cameras, they can detect structural cracks, leaks or overheating components without putting human workers in danger.

3. Search and Rescue operations

Emergency services deploy drones equipped with thermal imaging cameras to locate missing hikers, disaster victims, or criminals in dark, dense, or dangerous terrain. They can cover vast areas much faster than ground search parties and can fly into hazardous zones eg. wildfires

4. Aerial Photography and filmmaking.

Drones have revolutionized the entertainment and media industry by making sweeping, cinematic bird's-eye-view short shots affordable and accessible. They are widely used in movies, television shows, real estate marketing, sporting events and wedding videography

5. Medical & commercial Delivery

Logistics companies and healthcare providers use specialized cargo drones to transport urgent medical supplies, such as blood samples, vaccines and prescription medicines to remote villages, traffic-congested cities, or disaster-stricken areas where ground transport is too slow

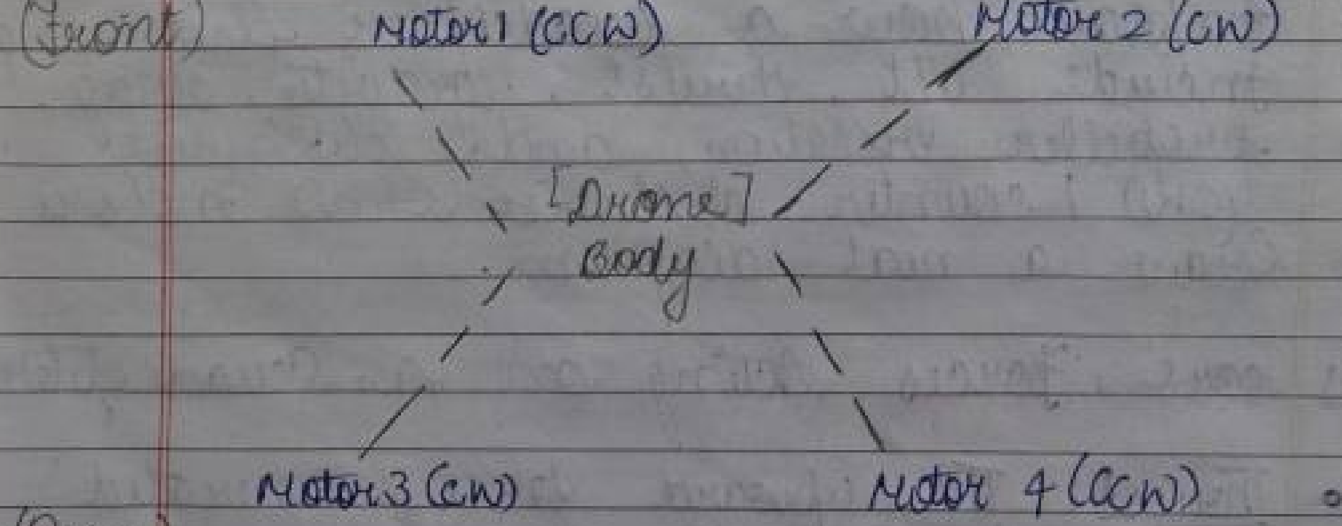
Q4 Explain how a quadcopter flies. & include lift, thrust, gravity, drag, propeller rotation, and clockwise (CW) / counter-clockwise (CCW) motors. Draw a neat diagram.

Ans core Forces acting on a Quadcopter

- Thrust: The upward force generated by the spinning propellers pushing air downward. According to Newton's Third Law of Motion as the propellers push air down an equal and opposite direction due to torque.

To countract this and maintain stability:

- A quadcopter uses four motors arranged in opposing pairs.
- Two diagonal motors spin clockwise
- The other two diagonal motors spin counter-clockwise.
- This opposing rotational ~~motors~~ cancels out the net rotational torque during a stable hover or flight.



→ Motor 1 & Motor 4 : counter - clockwise (CCW) rotation pair

→ Motor 2 & Motor 3 : clockwise (CW) rotation pair

→ YAW (Rotation control) : By increasing the speed of one diagonal pair (eg, the CW pair) while decreasing the speed of the CCW pair, the balance of torque is disrupted, causing the quadcopter to rotate cleanly to the left or right (yaw) without changing its position

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Q5 Explain how a drone performs take - off, landing, hovering, move forward, move left, move right, and rotate (YAW) . use suitable diagrams

sol Here is how a drone performs different movements

• Take off : All four motors spin faster to generate more upward thrust than gravity, pushing the drone into the air

- **Landing**: All four motors slow down so that gravity pulls the drone gently back to the ground.

- **Hovering**: All four motors spin at a balanced speed where total lift equals gravity keeping the drone steady in one place.

- **Move forward**: The rear motors speed up and the front motors slows down, tilting the drone forward and pushing it ahead.

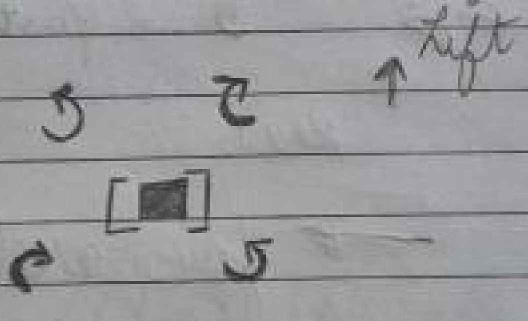
- **Move Left / Right**

→ To move left, the right-side motors speed up, tilting the drone to the left.

→ To move right, the left-side motors speed up, tilting the drone to the right.

- Rotate (YAW) changing the speed-balance between the (CW) and (CCW) motor pairs makes the drone spin left or right on its vertical axis without tilting.

Take off

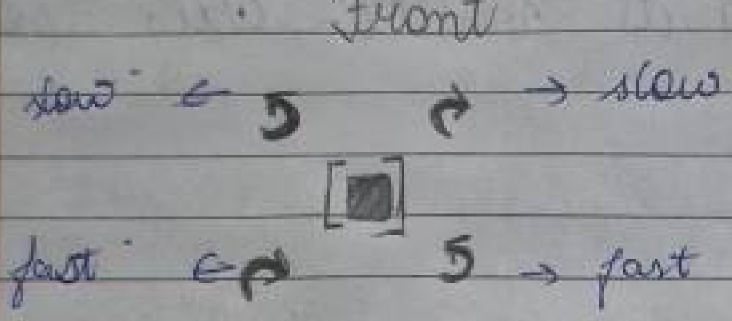


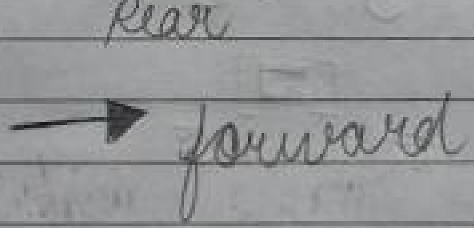
(all motors : fast)

Landing

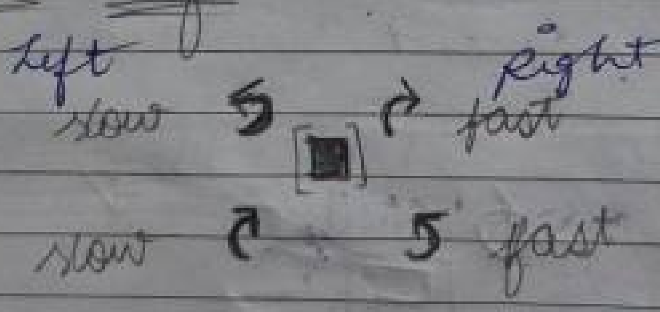
(all motors : slow)

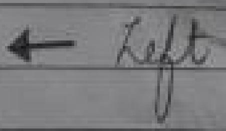
Move forward





Move left





Move Right

Left

Right

fast slow

fast slow

Right →

Movemint YAW (rotate)

fast slow

slow fast

YAW (rotate)

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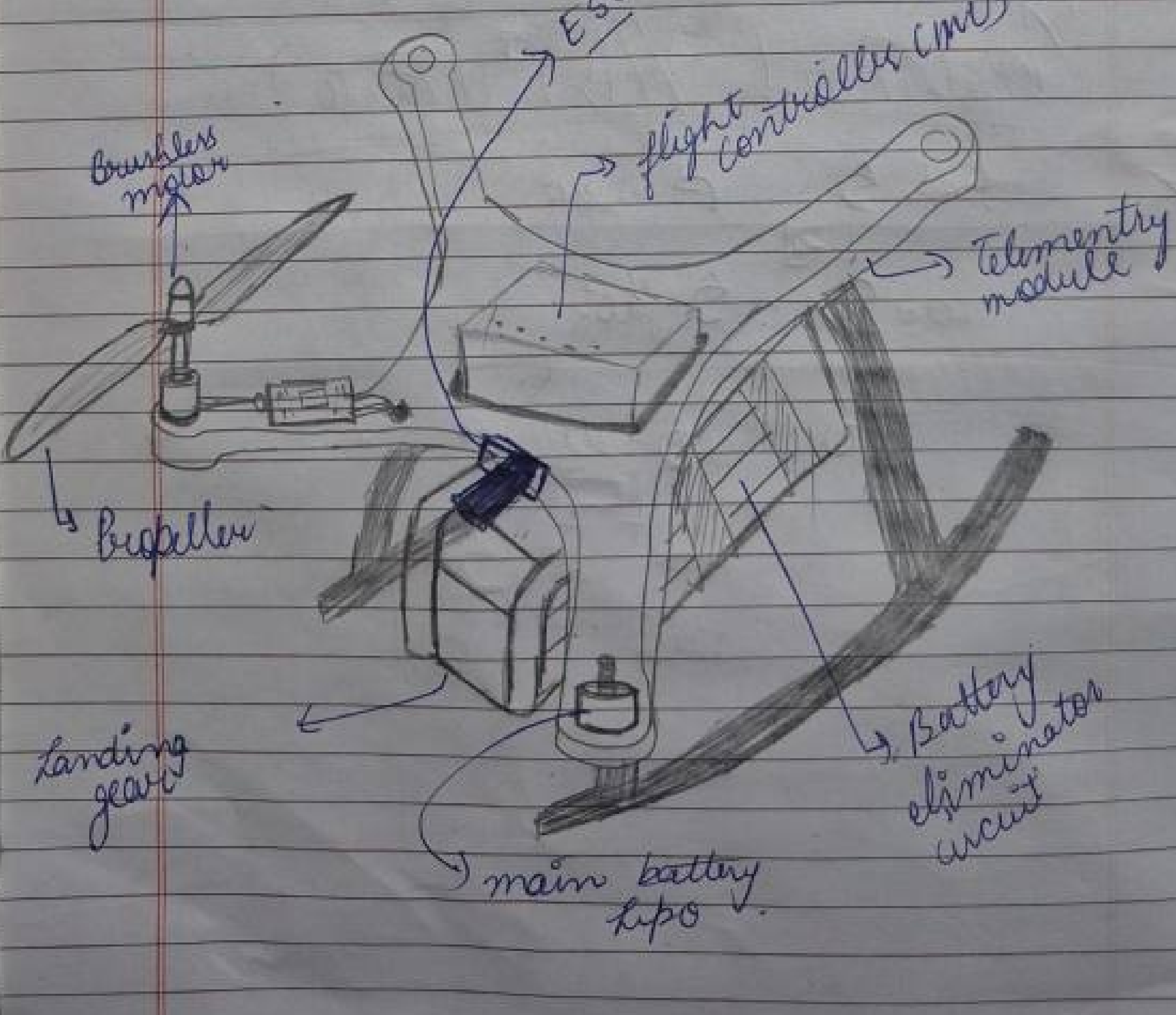
Q6. Draw a labled block diagram of a drone and explain the function of each component:
flight controller
ESC
brushless motors,
propillers
battery eleminator circuit (BEC)
telementary module



Functions of each component

1. Flight controller

→ It is brain of the drone
→ Receives commands from the remote and sensor data
→ controls the speed of each motor to stabilize and navigate the drone.

2. ESC (electronic speed controller)

→ controls the speed of the brushless motors
→ converts signals from the flight controller into motor movement

3. Brushless Motors

→ converts electrical energy into mech rotation

→ spin the propellers to generate lift

4. Propeller

→ Push air downward to create upward thrust

→ Help the drone take off, land, and move in diff direction

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5. Battery elimination circuit (BEC)

→ supplies a regulated low voltage (usually 5V) to the flight controller and other electronics

→ eliminates the need for a separate battery for these components.

6. Telementry Module

→ sends real-time flight data such as altitude, speed, battery level and GPS position to the ground station or remote controller

→ Helps monitor the drone during flight

Q7. Explain the role of the following components : flight controller, ESC, BLDC motor, ~~p~~ propeller, Li-po battery, GPS, Radio transmitter, frame, battery eliminator circuit, Radio receiver and frame

→ BLDC Motor (Brushless Direct current motor)

converts electrical energy into high speed rotational mechanical energy to spin the rotors efficiently

→ Li-po Battery (Lithium polymer Battery)

serves as the main power source supplying high-capacity electrical energy to spin the motors and electronics

→ GPS

Provides accurate geographic location, velocity and altitude tracking data for autonomous navigation hovering, and return-to home features

→ Radio Transmitter / Receiver

The transmitter (held by the pilot) sends control commands, while the onboard receiver catches those signals and passes them to flight controller

→ **Frame** :- The structural skeleton (Often made of Carbon fiber) that holds all components securely together while keeping weight to a minimum.

→ **Battery eliminator circuit (BEC)**:

→ sol 6.

Q8. explain the working principle and application of IMU, accelerometer, gyroscope, magnetometer, barometer, GPS, optical flow sensor, ultrasonic sensor, LIDAR, and camera. For each sensor mention purpose, working principle and application

sol IMU (Inertial Measurement unit)

Purpose → measure motion, orientation, and gravitational forces

working principle → combines multiple sensors to track changes in position and rotation over time.

Application used : used in drones, robotics and smartphones for balance and navigation.

2) Accelerometer

Purpose → measures linear acceleration and tilt

working principle → senses changes in ~~the~~ force on a microscopic internal mass, translating movement into electrical signals.

application → used in screen rotation on phones and fitness trackers to count steps.

3) ~~Gyr~~ Gyroscope

Purpose → Measures rotational speed and orientation

working principle → uses the principle of angular momentum (often via vibrating structures) to detect twisting or turning movements.

Application → used in gaming controllers camera stabilization and aircraft navigation

4 Magnetometer

Purpose :- measures magnetic fields and direction

working principle :- Detects the strength and direction of the earth's magnetic field, acting like a digital compass.

Application :- used in digital compass apps and indoor navigation systems.

5. Barometer

Purpose :- measures atmospheric pressure and altitude changes

working principle :- senses changes in air pressure, higher altitude equals lower pressure.

Application :- used in weather forecasting and altimeters in drones/smartwatches to track floors climbed.

6. GPS (global positioning system)

Purpose: determines precise geographical location and time.

working principle: receives radio signals from multiple satellites to calculate exact latitude, longitude, and altitude via triangulation.

application: used in vehicle navigation (google maps) and outdoor tracking

7. optical flow sensor

Purpose: measures relative motion against a surface

working principle: Takes rapid sequential images of the ground and analyzes pixel movement to determine horizontal speed and displacement.

application: used in indoor drones to maintain stable hovering without GPS

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8. Ultrasonic Sensor

Purpose → measures distance to nearby objects

Working principle : sends out a high frequency sound wave and measures the time it takes for the echo to bounce back from an obstacle

Application : used in car parking sensors and obstacle avoidance robotics

9. LiDAR (Light Detection and Ranging)

Purpose : create high-resolution 3D maps and precise distance measurements.

Working principle : emits rapid laser pulses and measures how long light takes to bounce back from objects (Time of flight)

Application :- used in ~~ad~~ autonomous self driving cars and 3D surveying

10 camera

Purpose :- captures visual images and video for environmental perception

working principle :- Focuses light through a lens onto an image sensor (CMOS/CCD) to convert Photons into digital pixel data

Application :- used in computer vision, security surveillance and autonomous vehicle lane detection

Q9. compare
a) GPS vs optical flow

GPS	optical flow
a) working principle uses satellite to determine location	a) WP uses downward facing camera and sensors to detect ground movement
b) Environment Best outdoors with clear sky.	b) Env Best indoors or at low altitude
c) Accuracy About 1-3 m.	c) very accurate at low heights
d) Altitude Range works at any altitude with satellite signal	d) effective up to about 10-20m depending on the sensor
e) signal req. Requires satellite signals	e) Requires visible ground texture and adequate lighting

b) LiDAR vs ultrasonic

LiDAR	Ultrasonic
a) W.P uses laser light pulses to measure distance	a) W.P uses high frequency sound waves to measure distance
b) Range Long (upto 100-300 m or more depending on model)	b) Range (short typically 2 cm to 4 5 cm)
c) Accuracy very high	c) Accuracy Moderate
d) speed very fast	d) speed very fast
e) Field of view Narrow and highly directional	e) Field of view wider beam angle
f) Power consumption Higher	f) Power consumpt Lower

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Q) Gyroscope vs Accelerometer

Gyroscope	Accelerometer
a) W.P Measures angular velocity	a) W.P. Measures linear acceleration and tilt
b) Measures Rotation around X, Y, Z axes	b) Measures acceleration along X, Y, Z axes
c) output angular velocity	c) output acceleration (m/s^2 or g)
d) Detects Roll, pitch, and yaw rotation	d) Movement vibration and tilt
e) Accuracy very accurate for short-term rotation	e) Accuracy good for detecting acceleration and tilt but affected by motion
f) Limitation suffers from drift over time	f) cannot accurately measure rotation alone

Q10 explain the concept of 6 Degree of Freedom (6DOF). Include translation, rotation, X-axis, Y-axis, Z-axis.. Draw suitable diagrams

sol **6DOF**

refers to the maximum number of independent ways a rigid body can move through three-dimensional space. It combines 3 translation movements (moving along axes) and 3 rotational movements (turning around axes).

1. Translation Movements (Linear motion)

Moving straight along the three physical axes without changing orientation.

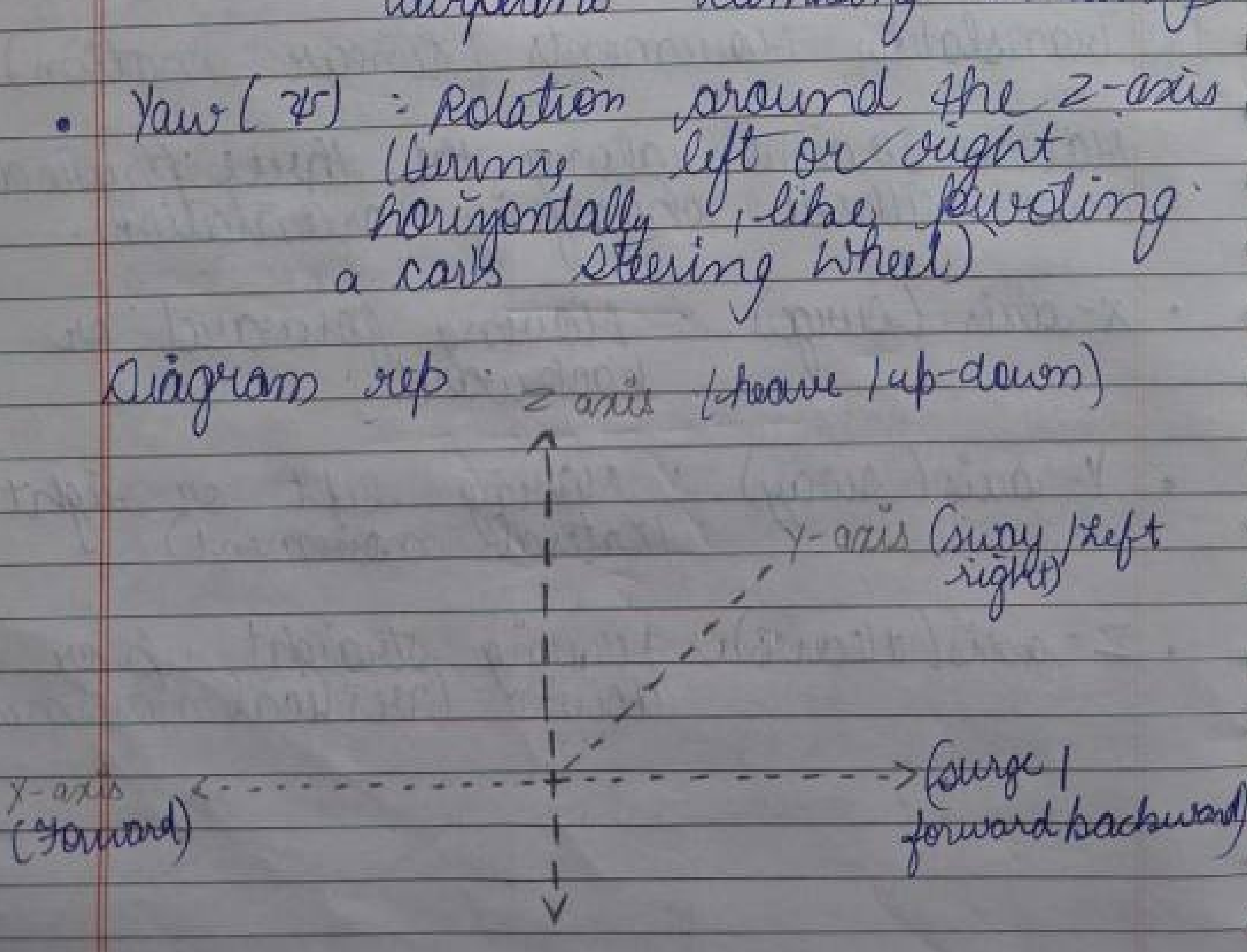
- X-axis (surge) : Moving forward or backward.
- Y-axis (sway) : Moving left or right (Lateral movement)
- Z-axis (Heave) : Moving straight up or down (Vertical movement)

2. Rotational movements (Angular Motion):

Tilting or turning around the three Phy axes:

- Roll (ϕ): Rotation around the X-axis. (tilting side-to-side, like banking a wing)
- Pitch (θ): Rotation around the Y-axis (tilting nose up or down, like an airplane climbing or diving)
- Yaw (ψ): Rotation around the Z-axis (turning left or right horizontally, like pivoting a car's steering wheel)

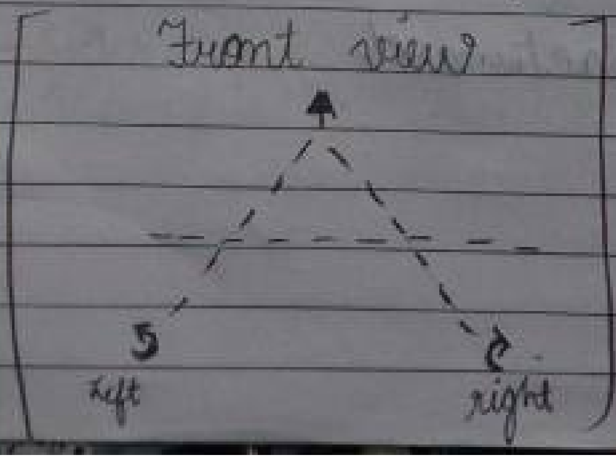
Diagram rep :



- Rotations
- Roll → Tilting sideways around the forward vector (X)
- Pitch → Tilting up/down around the lateral vector (Y)
- YAW → Turning left/right around the vertical vector (Z)

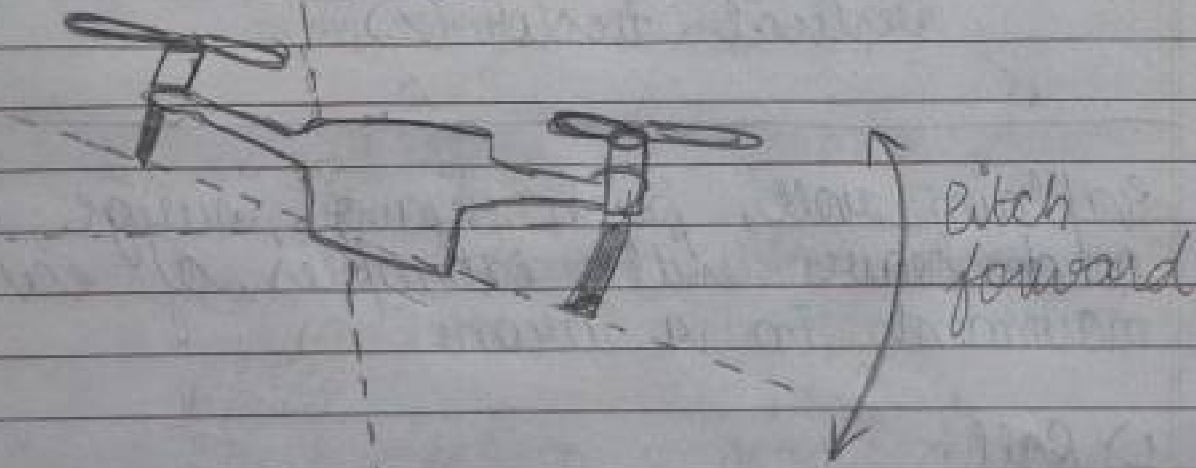
Q11 explain roll, pitch, yaw, surge, sway, and heave with examples of each movement in a drone

sol 1) Roll
- Definition : Rotation about the X-axis.
- Movement : The drone tilts left or right
- example : When the drone banks to the left or right while turning

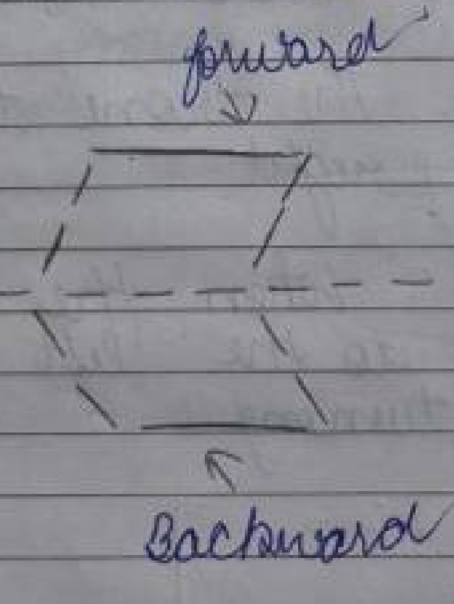


2. Pitch

- Definition : Rotation about Y-axis
- Movement : The drone tilts forward or backwards
- Example : Tilting forward to move ahead

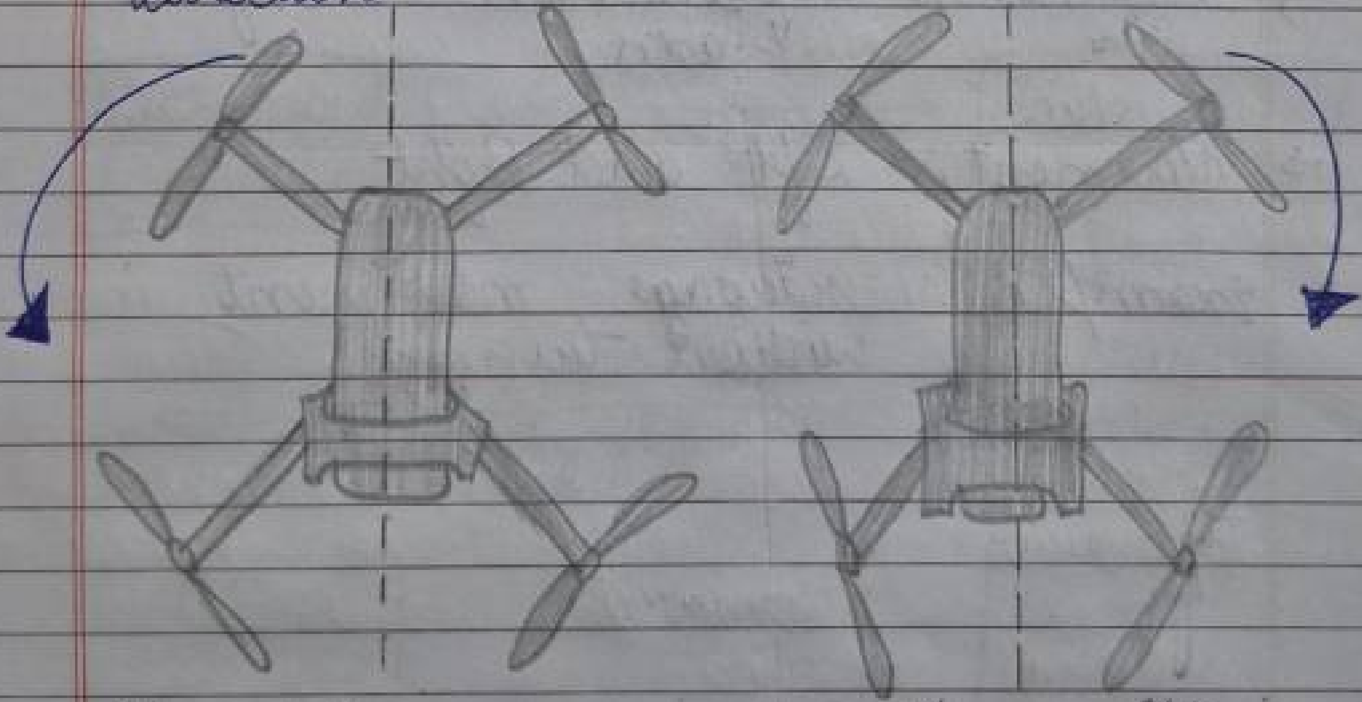


side view



3. yaw
→ Def : Rotation about the Z-axis.
→ Movement: The drone rotates left or right without changing its position.

→ Example:
turning the drone to face a diff. direction



Yaw left
(counter clockwise).

Yaw right
(clockwise).

4. Surge
• Definition : Translation along the X-axis.
• Movement : Forward and backward
• example : Flying straight forward or reversing

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← backward — Drone — Forward →

5) sway
→ Definition : Translation along the Y-axis

→ Movement : Left and Right

→ Example : sideways movement without turning

Top view

Forward

Left ← [Drone] → Right

6) Heave
→ Def : Translation along the Z-axis
→ Movement : up & down
→ example → Take off landing.

up
↑
[Drone]
↓
down

Q17. explain the complete signal flow in a drone
→ Pilot input to Radio transmitter to receiver to flight controller to ESC to BLDC motors to propellers to Drone movement. Illustrate with a block diagram.

1. Pilot Input :→ The pilot moves the control sticks or switches on the remote controller

2. Radio Transmitter
converts the Physical stick movement into the wireless radio signals and transmitts them into the air.

3. Receiver (on Drone).
captures the wireless Radio signals from the transmitter and converts them back into electrical control data signals

4. Flight controller:
Acts as the "brain" of the drone. It reads data from the receiver and onboard sensors, processes it and decides how much power or speed each motor needs to maintain stability or change direction

5. ESC.
Receives commands from the flight controller and regulates the precise amount of electrical power sent to the motors.

6. BLDC motor
converts electrical energy from the ESC into high speed rotational mech energy

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7 Propellers: spun by the motors to push air downward, generating the thrust and lift needed for flight

8. Drone Movement:

The net result of thrust, lift and torque changes makes the drone move, hover or turn in the air.

Q13 Why is PID control important in drones? Explain the proportional, integral and derivative terms

Sol PID control in Drones

Def: A PID (proportional-integral-Derivative) controller is a feedback control system used in drones to maintain stable flight by continuously correcting between the desired position / orientation and the actual position orientation

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Why is PID important?
→ Maintains stable flight.
→ Keeps the drone balanced during hover.
→ corrects errors caused by wind or external disturbances
→ ensures smooth and accurate movement.
→ Improve control of roll, pitch, yaw and altitude
→ Reduces oscillations and overshoot

components of PID controller.

1. Proportional (P)

function ÷ Produces an output proportional to the current error.

formula. $P = K_p \times e(t)$.

effect : The larger the error the stronger the correction

example : If the drone tilts 10° to the right the prop controller immediately commands the motors to level it

2) Integral-(I)

function : eliminates accumulated (steady-state) error over time.

formula

$$I = Ki \int e(t)\,dt$$

effect

corrects small errors that remain after proportional control.

example If the drone slowly drifts due to a constant wind, the integral term removes this drift.

3) Derivative (D)

Function : predicts future error by measuring how fast the error is changing.

Formula :

$$D = Kd \frac{de(t)}{dt}$$

effect Reduces overshoot and oscillations making the drone stable.

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Example:- If a drone is tilting too quickly the derivative term slows the correction to prevent excessive movement

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Q14. Explain obstacle avoidance in drones

sol obstacle avoidance is a feature that enables a drone to detect and avoid obstacles automatically during flight using sensors and intelligent control algorithms

Working:

1. sensor (Lidar, ultrasonic, Infrared, camera, radar) detect nearby obstacles.

2. The flight controller processes the sensor data.

3. If an obstacle is detected, the drone stops, slows down, or changes its path to avoid a collision.

Advantages:-

- → prevents crashes and damages
- → improves flight safety.
- → enables autonomous navigation
- → useful for mapping, surveillance and delivery drones

simple block diagram

obstacle
↓
[sensors]
↓
flight controller
↓
Drone changes direction

applications

- → Delivery drones
- → Agricultural drones.
- → surveillance and security.
- → search and rescue mission

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* Higher order Thinking Questions

1. Why do quadcopters require four motors instead of one?

sol → A single motor would make the whole drone spin in circles because of Physics.
Four motors spinning in opposite directions cancel out that ~~tw~~ twist so it stays steady.

→ To tilt or turn, the drone just speeds up or slows down specific motors. Doing this with just one motor would need super complicated and heavy mechanical parts

2. What would happen if one motor stops working during flight?

sol Loss of balance:
The drone will tilt or flip toward the side where the motor stopped because that side loses lift.

Uncontrolled spinning:
The balance of twisting forces (torque) gets ruined, causing the drone

to spin widly in circles.

→ crash:
unless it has a smart flight controller that instantly adjusts the remaining motors to compensate, the drone will quickly lose control and crash

3. Why are brushless motors preferred over brushed motors?

sol Longer life: They have no physical carbon brushes to wear down, so they last much longer.

→ More efficient as they create less friction and heat and can save battery power

→ They don't need regular cleaning

4. can a drone fly indoors without GPS? explain

sol Yes a Drone can easily fly indoors.

Instead GPS they use optical flow sensors to track ground, infrared ultrasonic sensors to measure distance from walls and floors, and IMU to maintain balance and stability

5. Why are Li-Po batteries used in drones instead of standard batteries?

sol High power : They deliver massive amounts of current instantly, which motors need to lift and fly fast.

Lightweight : They are much lighter than standard batteries, keeping the drone's payload low.

Flexible shapes and are lightweight

6. What factors reduce a drone's flight time

sol . extra weight or cameras forces motors to work harder, draining the battery faster

Flying against strong winds.

or in cold temperature decreases battery efficiency.

Fast speed and sharp turns also consume power

7. How can drone stability be improves during strong winds-

sol → using a slightly heavier drone
→ use smaller blades or small propellers creates less surface area for the wind to catch
→ Turning the PID settings allows the drone's brain to react faster to sudden error

8. Why is sensor fusion important in drones.

sol → It merges information from multiple sensors to give one accurate position

→ Individual sensors can be noisy or prone to drifting; combining them cancels out those errors.

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9. explain how GPS and IMU work together for stable navigation

sol GPS → provides global position and route tracking, but updates slowly.

IMU → Tracks fast movements tilts and rotations instantly

Teamwork → GPS keeps the drone on the right path, while IMU handles instant balance b/w GPS updates.

10. If you were designing a drone for mountain rescue operations, which sensors and components would you choose and Why?

sol Thermal camera
To detect body heat of missing people in snow or dark forests

GPS ~~or~~ & Barometer
For high-altitude navigation and tracking exact positions in ~~sensor~~ remote areas.

LiDar / obstacle sensors
To safely avoid crashing into the trees, rocks and cliffs in tight spaces